



भारतीय प्रबंध संस्थान कोषिकोड
Indian Institute
of Management
Kozhikode

Globalizing Indian Thought

WEEK 14

CAPSTONE ASSIGNMENT

AI-POWERED SMART ENERGY MANAGEMENT SYSTEM (SEMS)

MVP

SUBMITTED BY

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AI-Powered Smart Energy Management System

Capstone project to optimize energy consumption and promote sustainability.
Uses AI for real-time insights and actionable recommendations.



Project Roadmap



1

Planning

Comprehensive requirements gathering and detailed planning documents completed by Month 2.

2

Data & Models

Develop initial prototype and commence data collection by Month 4, focusing on relevant datasets.

3

Integration

Complete AI system integration and rigorous testing by Month 6 to ensure seamless functionality.

4

Deployment

Scale system for production deployment by Month 12, ensuring robust performance and reliability.

Risk Analysis & Mitigation

Data Privacy

Encrypted communication and GDPR compliance.

IoT Integration

Partner with reliable vendors and conduct integration tests.

High Costs

Explore partnerships and government subsidies.



AI Techniques

1

Demand Forecasting

ARIMA, Prophet, and LSTM models predict energy needs.

2

Energy Optimization

Reinforcement learning adjusts consumption in real-time.

3

Anomaly Detection

Isolation Forest and Autoencoders identify unusual patterns.

https://gamma.app/?utm_source=made-with-gamma

Data Collection Strategy



Historical Data

Smart meters provide consumption data.



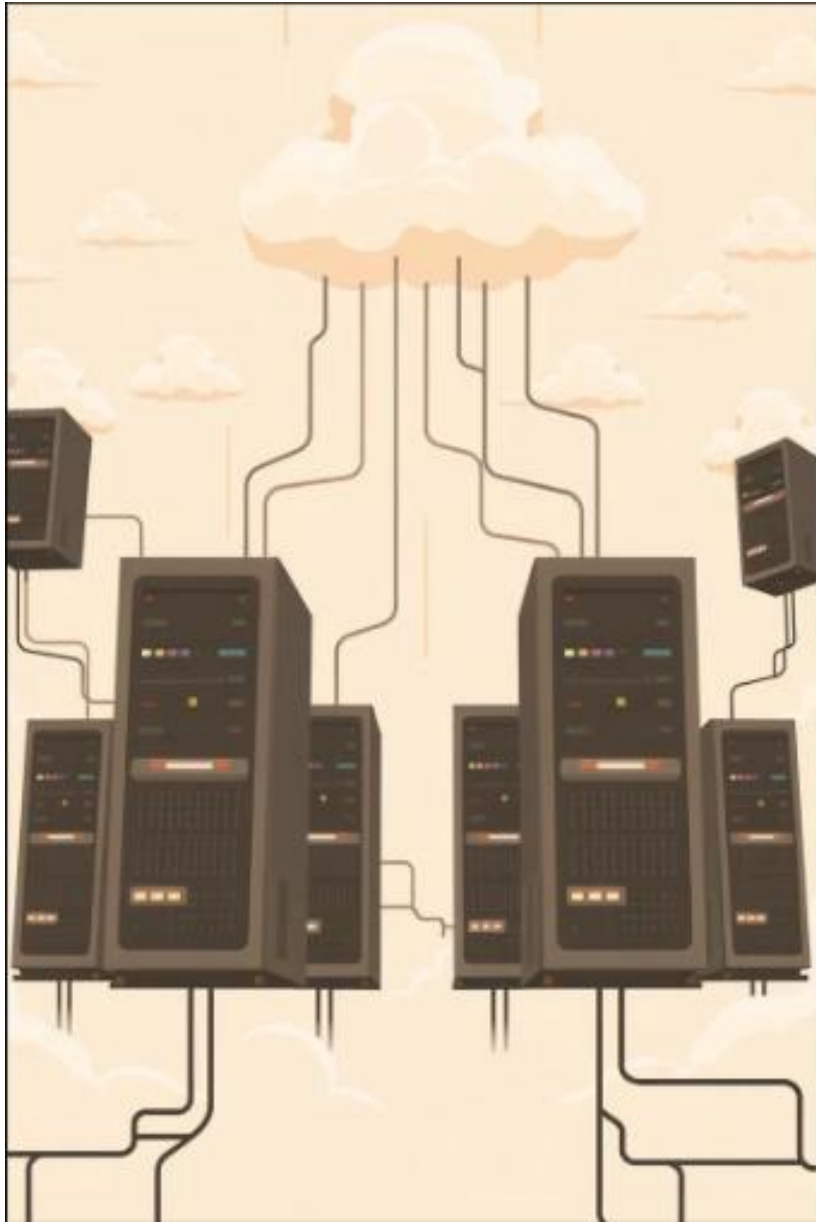
Weather Data

External sources improve forecasting.



Smart Appliances

Connected devices monitor usage.



Technology Infrastructure

IoT Devices

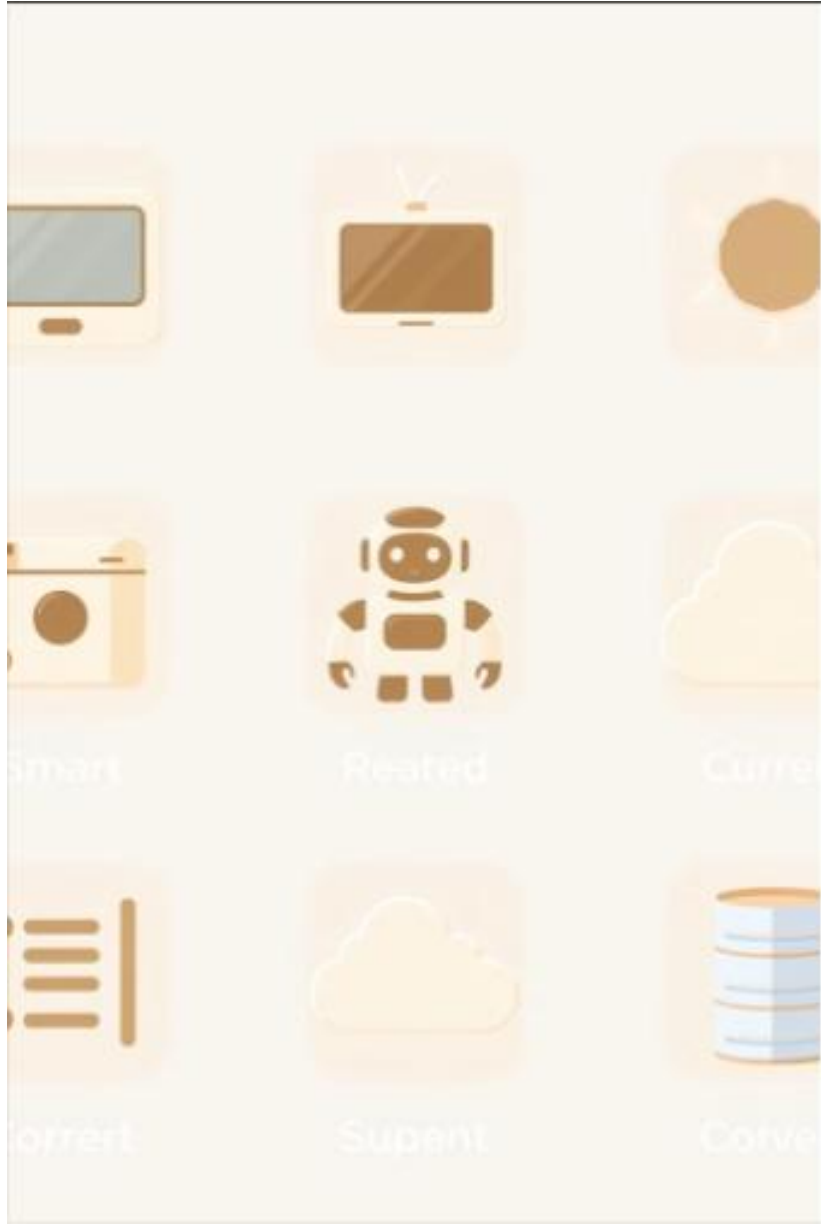
Smart meters for real-time monitoring.

AI Infrastructure

Cloud and edge computing for processing.

Data Security

Encrypted protocols and GDPR compliance.



Build vs. Buy

- 1 **IoT Devices**
Buy from reputable vendors.
- 2 **AI Models**
Custom-build for tailored solutions.
- 3 **Cloud**
Purchase cloud infrastructure.

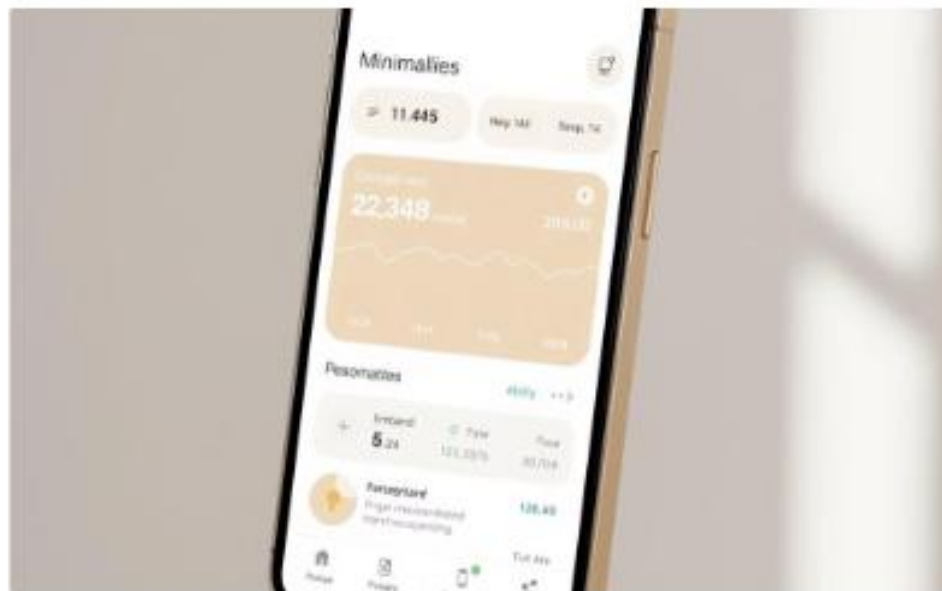


System Integration



Seamless integration ensures real-time monitoring and optimization.

MVP Design



Mobile App

Real-time data and recommendations.



Dashboard

Detailed analytics and alerts.



Conclusion

SEMS optimizes energy usage and promotes sustainability. The MVP provides essential features and scalable infrastructure.